

[illegible]

SPRING AI

GENERATIVE ARTIFICIAL INTELLIGENCE CON JAVA

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Corso avanzato per Venis S.p.A, Venezia, Italia

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Note

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SPEECH-TO-TEXT

Note

This image shows a single sheet of white paper with horizontal blue or grey ruling lines. The lines are evenly spaced and run across the width of the page. There are approximately 20 lines visible. The paper has a slight shadow on the right side, suggesting it's resting on a surface.

- Simone Scannapieco

Note

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```

...
<dependency>
  <groupId>org.springframework.boot</groupId>
  <artifactId>spring-boot-starter-web</artifactId>
</dependency>
<dependency>
  <groupId>org.springframework.ai</groupId>
  <artifactId>spring-ai-starter-model-openai</artifactId>
</dependency>
...

```

```
spring:
  application:
    name: demo
  ai:
    openai:
      api-key: ${OPENAI_API_KEY}
```

Note

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```
@RestController
@RequestMapping("/api")
public class AudioController {

    private final OpenAiAudioTranscriptionModel openAiAudioTranscriptionModel;

    AudioController(OpenAiAudioTranscriptionModel transcriptionModel,
        SpeechModel speechModel) {
        this.openAiAudioTranscriptionModel = transcriptionModel;
        this.speechModel = speechModel;
    }

    @GetMapping("/transcribe")
    String transcribe(@Value("classpath:filename.mp3") Resource audioFile) {
        return openAiAudioTranscriptionModel.call(audioFile);
    }
}
```

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- 1 Modifica configurazione di sistema
- 2 Creazione interfaccia ed implementazione del servizio
- 3 Modifica controllore MVC
- 4 Test delle funzionalità con Postman/Insomnia

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Configurazione applicativo

```

spring:
  application:
    name: demo
  ...
  ai:
    ...
    openai:
      ...
      audio:
        transcription:
          options:
            model: gemini-2.5-flash
  
```

Note

Note-taking area with horizontal lines.

```
package it.venis.ai.spring.demo.services;

import org.springframework.core.io.Resource;

import it.venis.ai.spring.demo.model.Answer;

public interface MultiModalityService {

    Answer getTranscriptionFromAudioFile(Resource audioFile);

}
```

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```
package it.venis.ai.spring.demo.services;

...

@Service
public class MultiModalityServiceImpl implements MultiModalityService {

    private final ChatClient geminiChatClient;

    public MultiModalityServiceImpl(@Qualifier("geminiChatClient") ChatClient geminiChatClient) {

        this.geminiChatClient = geminiChatClient;

    }

    @Override
    public Answer getTranscriptionFromAudioFile(Resource audioFile) {

        String contentType = getContentType(audioFile);

        Media audioMedia = new Media(
            MimeTypeUtils.parseMimeType(contentType), audioFile
        );

        return new Answer(this.geminiChatClient.prompt()
            .user(u -> u.text("Trascrivi il seguente file audio: ").media(audioMedia))
            .call()
            .content()
        );

    }

    ...
}
```

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```
...
private String getContentType(Resource audioFile) {
    String contentType = URLConnection.guessContentTypeFromName(audioFile.getFilename());

    if (contentType == null || contentType.equals("application/octet-stream")) {
        String fileName = audioFile.getFilename();
        if (fileName != null) {
            String extension = fileName.substring(fileName.lastIndexOf('.') + 1).toLowerCase();
            contentType = switch (extension) {
                case "mp3" -> "audio/mpeg";
                case "wav" -> "audio/wav";
                case "m4a" -> "audio/mp4";
                case "ogg" -> "audio/ogg";
                case "flac" -> "audio/flac";
                default -> "audio/mpeg";
            };
        } else {
            contentType = "audio/mpeg";
        }
    }

    return contentType;
}
}
```

Note

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```
package it.venis.ai.spring.demo.controllers;

...

@RestController
@Configuration
public class QuestionController {

    ...

    private final MultiModalityService multiModalityService;
    ...

    public QuestionController(QuestionService service,
        RAGService ragService,
        TimeToolsService timeToolsService,
        MultiModalityService multiModalityService,
        ...) {

        ...
        this.multiModalityService = multiModalityService;
        ...

    }

    ...

    @PostMapping("/gemini/ask/multi-modality/transcribe")
    public Answer getTranscriptionFromAudioFile(@Value("classpath:Venis_descrizione_azienza.wav") Resource audioFile) {
        return this.multiModalityService.getTranscriptionFromAudioFile(audioFile);
    }

}
```

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Branch:

16-spring-ai-gemini-ollama-multimodality-audio-transcription

Note

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